DevelopersHub Corporation

AI/ML Engineering Internship Tasks

Due Date: 26th June, 2026

Overview

As part of your AI/ML Engineering Internship at DevelopersHub Corporation, you are required to complete **at least 3 out of the following 6 tasks**. These tasks are designed to help you build a strong foundation in artificial intelligence and machine learning through real-world datasets and use cases.

You will practice skills like data preprocessing, model training, evaluation, prompt engineering, and even chatbot development — all of which are essential in AI/ML careers.

Task 1: Exploring and Visualizing a Simple Dataset

Objective:
Learn how to load, inspect, and visualize a dataset to understand data trends and distributions.

Dataset:
Iris Dataset (CSV format, can be loaded via seaborn or downloaded)

Instructions:

- Load the dataset using pandas.
- Print the shape, column names, and the first few rows using .head().
- Use .info() and .describe() for summary statistics.
- Visualize the dataset:
  - Create a scatter plot to show relationships between features.

    ○ Use histograms to show value distributions.

    ○ Use box plots to identify outliers.

- Use matplotlib and seaborn for plotting.

Skills:

- Data loading and inspection using pandas
- Descriptive statistics and data exploration
- Basic plotting and visualization with seaborn and matplotlib

Task 2: Predict Future Stock Prices (Short-Term)

Objective:
Use historical stock data to predict the next day's closing price.

Dataset:
Stock market data from Yahoo Finance (retrieved using the yfinance Python library)

Instructions:

- Select a stock (e.g., Apple, Tesla).
- Load historical data using the yfinance library.
- Use features like Open, High, Low, and Volume to predict the next Close price.
- Train a Linear Regression or Random Forest model.
- Plot actual vs predicted closing prices for comparison.

Skills:

- Time series data handling
- Regression modeling

- Data fetching using APIs (yfinance)
- Plotting predictions vs real data

Task 3: Heart Disease Prediction

Objective:
Build a model to predict whether a person is at risk of heart disease based on their health data.

Dataset:
Heart Disease UCI Dataset (available on Kaggle)

Instructions:

- Clean the dataset (handle missing values if any).
- Perform Exploratory Data Analysis (EDA) to understand trends.
- Train a classification model (Logistic Regression or Decision Tree).
- Evaluate using metrics: accuracy, ROC curve, and confusion matrix.
- Highlight important features affecting prediction.

Skills:

- Binary classification
- Medical data understanding and interpretation
- Model evaluation using ROC-AUC and confusion matrix
- Feature importance analysis

Task 4: General Health Query Chatbot (Prompt Engineering Based)

Objective:
Create a chatbot that can answer general health-related questions using an LLM (Large Language Model).

Tools:
OpenAI GPT-3.5 via API (or use a free open-source model like Mistral-7B-Instruct on Hugging Face)

Instructions:

- Build a simple Python script or notebook that sends user queries to an LLM.
- Use prompt engineering to make the responses friendly and clear (e.g., "Act like a helpful medical assistant").
- Add safety filters to avoid giving medical advice that could be harmful.
- Example queries:
    - "What causes a sore throat?"
    - "Is paracetamol safe for children?"

Skills:

- Prompt design and testing
- Using APIs for LLMs (e.g., OpenAI, Hugging Face)
- Safety handling in chatbot responses
- Building simple conversational agents

Task 5: Mental Health Support Chatbot (Fine-Tuned)

Objective:

Build a basic chatbot that provides supportive and empathetic responses for stress, anxiety, and emotional wellness.

Model Base:
DistilGPT2, GPT-Neo, or Mistral (7B)
Dataset for Fine-Tuning:
EmpatheticDialogues (Facebook AI)

Instructions:

- Fine-tune a small LLM using Hugging Face's Trainer API.
- Train it to respond empathetically using real human dialogues.
- Ensure the tone is gentle and emotionally supportive.
- Build a command-line or Streamlit-based interface to test it.

Skills:

- Model fine-tuning with Hugging Face Transformers
- Emotional tone design for safe chatbots
- Conversation modeling
- Deployment using CLI or simple web apps

Task 6: House Price Prediction

Objective:
Predict house prices using property features such as size, bedrooms, and location.

Dataset:
House Price Prediction Dataset (available on Kaggle)

Instructions:

- Perform preprocessing on features like square footage, number of bedrooms, and location.

- Train a regression model (Linear Regression or Gradient Boosting).
- Visualize predicted prices compared to actual prices.
- Evaluate with Mean Absolute Error (MAE) and RMSE.

Skills:

- Regression modeling
- Feature scaling and selection
- Model evaluation (MAE, RMSE)
- Real estate data understanding

Submission Requirements (Checklist for Each Task)

To receive credit for a task, ensure the following:

1. Jupyter Notebook

- Clear problem statement and goal
- Dataset loading and preprocessing
- Data visualization and exploration
- Model training and evaluation
- Explanation of results and final insights

2. Code Quality

- Code should be clean, modular, and commented
- Include explanations of each major step

3. GitHub Repository

- Create a GitHub repo for your AI/ML internship tasks
- Add a README.md file summarizing:
  - Task objective
  - Dataset used
  - Models applied
  - Key results and findings

4. Submission on Google Classroom

- Share your GitHub repository link for each completed task.

Important Note

- You are required to complete **at least 3 out of the 6 tasks** before the due date: **26th June, 2026**.
- You may complete more tasks for additional learning and project portfolio strength.
- If you need help, feel free to ask questions in the group or consult with your mentors.